

MINERVA CryoCell P&ID — Default Views Documentation (v4)

Project: MINERVA CryoCell — SCK CEN (MYRRHA/MINERVA Phase 1)

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Standard: SCK CEN AD_01.16 · Status: S2 — FOR ACCEPTANCE · RESTRICTED

1. Purpose

The v4 P&ID set is **layered**. Each SVG contains 24 named layers (groups), and the drawing ships with five pre-defined **default views** — curated layer combinations for different audiences and tasks. The view definitions are embedded in each SVG's `<metadata>` block and mirrored in the interactive viewer (`js/pid-viewer.js` → `VIEWES`), so the same presets are available whether you open the SVG directly or use the HTML viewer.

2. The 24 Layers (draw order, bottom → top)

Layers draw in list order — the first entry is the bottom of the stack. Tag layer `12_Tags` and overlays are placed late so they render on top of pipes and equipment.

Grouped by function:

- **Frame / title:** `00_Frame` , `01_TitleBlock`
- **Process pipes (by class):** `02_Pipe_D` , `02B_Pipe_E` , `03_Pipe_A` , `04_Pipe_B` , `04C_Pipe_WATER` , `05_Pipe_AIR`
- **Equipment:** `06_Equipment` , `07_Vessels` , `08_Valves`
- **Overlays:** `08B_Valves_HORIZONTAL_OVERLAY` (hidden by default)
- **Instruments / signals:** `09_Instruments` , `10_Signals` , `11_Controls`
- **Annotation:** `12_Tags` , `13_LineNames` , `14_TerminalPoints` , `15_Dimensions`
- **Reference:** `16_Legend` , `16_Legend_INTERACTIVE` (hidden), `17_Notes_TOGGLEABLE` (hidden), plus revision/notes layers.

Three layers are **hidden by default**:

`08B_Valves_HORIZONTAL_OVERLAY` , `16_Legend_INTERACTIVE` , `17_Notes_TOGGLEABLE` .

3. The Five Default Views

View	Audience / use case	Shows
FULL	Complete reference / acceptance review	All visible layers (everything except the 3 hidden-by-default).
PROCESS	Process engineers — focus on fluid path	Frame, title, all process-pipe classes, equipment, vessels, valves, line names, terminal points, legend. Hides most instrument/signal detail.
CONTROL	C&I engineers — focus on instrumentation	Frame, title, equipment outlines, instruments, signals, controls, tags, legend. Pipes shown light for context.
MAIN	General overview / drawing issue	Frame, title, primary process pipes, equipment, vessels, valves, tags, line names, terminal points, legend.
PRINT_MONO	Black-and-white plotting / AD_01.16 mono check	All process content in monochrome with inline line names and weight/dash differentiation; colour suppressed.

View details

- **FULL** — `vis = layer not in HIDDEN_BY_DEFAULT` . The horizontal-valve overlay, interactive legend and toggleable notes remain off until explicitly enabled.
- **PROCESS** — emphasises `02*-05*` pipe classes, `06/07/08` equipment & valves, `13_LineNames` , `14_TerminalPoints` . Instrument layers (`09/10/11`) are off.
- **CONTROL** — emphasises `09_Instruments` , `10_Signals` , `11_Controls` , `12_Tags` ; process pipes kept visible but visually subordinate for context.
- **MAIN** — the “issue” view: primary headers + equipment + valves + tags + names.
- **PRINT_MONO** — drives the mono rendering path so a colour-free A3 plot still conveys every class via names, weights and dash patterns (see Colour & Legend Guide).

4. Using the Views

- **In the SVG:** the `<metadata id="minerva-frontmatter">` block lists each view and its layer visibility map (machine-readable, inside a CDATA section so the YAML `---` fences don’t break XML parsers).

- **In the HTML viewer:** select a preset from the **Views** control; the layer panel updates to reflect which layers each preset enables. You can then fine-tune individual layers on top of any preset.

See `INTERACTIVE_VIEWER_MANUAL.md` for step-by-step viewer instructions.